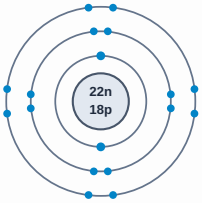
STANDARD IX — CHEMISTRY (ANSWER KEY)

First Terminal Examination 2026 | EM

Subject: Chemistry | Class: 9 | Total Marks: 40

Q. No.	Value Points / Expected Answers	Marks
1	Atomic number **(b)**	1
2	A is correct, R is incorrect **(a)**	1
3	Statements i & iii **(c)**	1
4	s, p, q **(c)**	1
5	**a)** Nucleus **b)** Positive charge	2
6	**a)** They have the same mass number. **b)** Isobar	2
7	**a)** Atomic number = **8** **b)** Electronic configuration = **2, 8**	2
8	**a)** Most of the space inside an atom is empty. **b)** The entire positive charge and almost the whole mass of the atom are concentrated in the nucleus.	2
9.A	• Arranged in the increasing (ascending) order of atomic numbers. • Elements with similar properties are placed in the same group.	2
9.B	**OR (Choice B)** • Could not assign a definite/proper position to Hydrogen. • Could not assign appropriate positions to Isotopes.	2
10	**a)** When an electric field was applied in the path of the rays, they deflected towards the positive plate. **b)** When a light paddle wheel was placed in their path, it rotated.	2
11.A	**R**. All of them belong to the same period. Moving from left to right across a period, the nuclear charge increases and the atomic size decreases.	2
11.B	**OR (Choice B)** **a)** x > y > z **b)** Atomic size increases because the number of shells increases down the group.	2
12.A	**a)** $_{8}O^{17}$, $_{8}O^{16}$ **b)** They have the same atomic number but different mass numbers (Isotopes). **c)** Na-24 (Sodium-24)	3
12.B	**OR (Choice B)** **a)** Neutron **b)** 3 **c)** Deuterium ($_{1}H^{2}$)	3

Q. No.	Value Points / Expected Answers	Marks
13	**a)** 13 **b)** 2, 8, 1 **c)** 9	3
14.A	**a)** 18 **b)** Bohr model of Argon atom: Nucleus: 18 Protons (18p), 22 Neutrons (22n) Electronic Configuration: 2, 8, 8 (K=2, L=8, M=8) 	3
14.B	**OR (Choice B)** **a)** Bohr atom model / Atomic structure representation (15 Protons, 16 Neutrons, 15 Electrons → Phosphorus: 2, 8, 5) **b)** K shell	3
15	**a)** L, N **b)** 19 **c)** N	3
16	**a)** **Y** — Because it contains 8 electrons in its outermost shell (stable octet configuration). **b)** 2	3
17	**a)** 3 **b)** Aluminium (Al) **c)** 8	3
18.A	**a)** In the shell immediately inner to the outermost shell (penultimate shell / inner shell). **b)** Transition elements. **c)** • Show variable oxidation states. • Form colored compounds.	4
18.B	**OR (Choice B)** **a)** Groups 1, 2 and 13 to 18 (Representative elements / Main group elements). **b)** 2 **c)** • Show similar chemical properties. • Makes the study of elements easier and systematic.	4